Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *NLLB* | ***Context:*** *Amhara Region Agricultural and Microfinance Translation Cohort*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's task taxonomy is organized along resource level (high/low/very low) and a narrow set of domains (news, scripted talks, chat, WHO health) tested only in NLLB-MD for six language directions [Q159, Q166-Q169], none of which include Amharic. The deployment's core document genres — agricultural input subsidy notices, land-use policy updates, and credit-scheme terms — fall entirely outside the evaluation taxonomy. The paper itself frames the generalization question as 'How well do models generalize to non-Wikimedia domains?' [Q159] but does not extend domain testing to administrative, agricultural, or financial genres. Independent 2025 evidence confirms Flores+ scores do not reliably predict out-of-domain translation quality [WEB-10]. Given the HIGH priority assigned to this dimension by the elicitation, the absence of any agricultural/bureaucratic/financial category in the benchmark taxonomy is a significant content-validity concern.

ID	Evidence
Q159	“More specifically, we want to answer the following two questions: (1) How well do models generalize to non-Wikimedia domains? (2) Does fine-tuning on high quality in-domain parallel text lead to good performance?” (p.24)
Q166	“We translate the English side of the WMT21 English-German development set, containing a sample of newspapers from 2020 (Akhbardeh et al., 2021).” (p.25)
Q167	“We translate text extracted from a series of scripted English-language talks covering a variety of topics.” (p.25)
Q168	“We extract 3,000 utterances from the multi-session chat dataset of Xu et al. (2022), which contains on average 23 words per turn.” (p.25)
Q169	“We translated one World Health Organisation report (Donaldson and Rutter, 2017) and combined it with sentences translated from the English portion of the TAUS Corona Crisis Report.” (p.25)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)

Strengths

- The benchmark explicitly anticipates the domain-generalization question and creates NLLB-MD to probe it [Q159], providing a documented framework (if not coverage) for thinking about non-Wikimedia domains.
- Resource-level stratification (high/low/very-low) [Q419, Q420] gives users a coarse signal of expected difficulty for Amharic, classified as low-resource.

ID	Evidence
Q159	“More specifically, we want to answer the following two questions: (1) How well do models generalize to non-Wikimedia domains? (2) Does fine-tuning on high quality in-domain parallel text lead to good performance?” (p.24)
Q419	“In terms of resource level, there are 40 high-resource and 70 low-resource directions (see Table 1 and Table 51).” (p.49)
Q420	“Out of 70 low-resource directions, 22 are very low-resource, i.e., have less than 100K training examples.” (p.49)

Checklist

Checklist item definitions:

ID	Assessment Question

IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required deployment categories include: agricultural input subsidy notices, land-use/land-tenure policy updates, microfinance credit-scheme terms, and bureaucratic/legal register documents with numbered clauses and conditional eligibility logic. None of these are represented in Flores-200 or NLLB-MD task taxonomy [Q95, Q166-Q169]; web search confirms no Ethiopian agricultural or administrative MT benchmark exists [WEB-3, WEB-4, WEB-10].

IO-2: Yes — the taxonomy omits all regionally relevant administrative/financial/agricultural categories. NLLB-MD covers only news, scripted talks, chat, and WHO health [Q166-Q169], and only for six languages not including Amharic [Q161]. The Flores-200 evaluation set is sourced from Wikinews, Wikijunior, and Wikivoyage [Q95].

IO-3: The benchmark includes ancillary subtasks (LID [Q179], bitext mining [Q286], toxicity [Q699]) that are upstream model-development tasks rather than irrelevant for the deployment per se, but no category is irrelevant in a way that consumes evaluation weight. Domain coverage in NLLB-MD (chat, scripted talks) is not aligned with the deployment but does not displace deployment-relevant evaluation since Amharic is excluded from NLLB-MD entirely [Q161].

IO-4: Category gaps that harm content validity: (a) no agricultural/subsidy genre, (b) no land-tenure/legal proclamation genre, (c) no microfinance/credit-agreement genre, (d) no document-level coherence or terminology-consistency task. These omissions are confirmed structural — not addressable by tuning thresholds [WEB-10 confirms domain-benchmark mismatch is real].

ID	Evidence
Q95	“As a significant extension of Flores-101, Flores-200 consists of 3001 sentences sampled from English-language Wikimedia projects for 204 total languages. Approximately one third of sentences are collected from each of these sources: Wikinews, Wikijunior, and Wikivoyage. The content is professionally translated into 200+ languages to create Flores-200. As we translate the same set of content into all languages, Flores-200 is a many-to-many multilingual benchmark.” (p.19)
Q159	“More specifically, we want to answer the following two questions: (1) How well do models generalize to non-Wikimedia domains? (2) Does fine-tuning on high quality in-domain parallel text lead to good performance?” (p.24)
Q161	“NLLB-MD covers the following six languages: Central Aymara (ayr_Latn), Bhojpuri (bho_Deva), Dyula (dyu_Latn), Friulian (fur_Latn), Russian (rus_Cyrl) and Wolof (wol_Latn).” (p.24)
Q166	“We translate the English side of the WMT21 English-German development set, containing a sample of newspapers from 2020 (Akhbardeh et al., 2021).” (p.25)
Q167	“We translate text extracted from a series of scripted English-language talks covering a variety of topics.” (p.25)
Q168	“We extract 3,000 utterances from the multi-session chat dataset of Xu et al. (2022), which contains on average 23 words per turn.” (p.25)
Q169	“We translated one World Health Organisation report (Donaldson and Rutter, 2017) and combined it with sentences translated from the English portion of the TAUS Corona Crisis Report.” (p.25)
Q179	“Language identification (LID) is the task of predicting the primary language for a span of texts.” (p.26)
Q286	“The underlying idea of our bitext mining approach is to first learn a multilingual sentence embedding space and to use a similarity measure in that space to decide whether two sentences are parallel or not.” (p.37)
Q699	“Our goal in this section is to provide an analysis of toxicity in multilingual MT models.” (p.79)
WEB-3	No Ethiopian agricultural or administrative-domain Amharic benchmark exists (https://arxiv.org/abs/2403.13737)
WEB-4	EthioMT parallel corpora (15 Ethiopian languages) drawn from religious and general domains, not administrative/financial (https://arxiv.org/html/2403.19365v1)
WEB-10	2025 study: mock translations merely reproducing named entities achieved non-trivial BLEU and ChrF++ scores, confirming metrics can reward superficial overlap rather than translational adequacy (https://arxiv.org/html/2508.20511v1)

Information Gaps

- Whether any private-sector or government-internal Amharic agricultural/financial MT benchmark exists; web search found none.

Requires Expert Verification

- Whether the document-genre distinctions matter as much in practice as the elicitation suggests, given that source documents are primarily already in Amharic (per regional context, the deployment is converting documents into Amharic).

Remediation

Gap 1: No agricultural, land-tenure, microfinance, or bureaucratic-legal genres in the task taxonomy; NLLB-MD covers only news/talks/chat/health and excludes Amharic [Q161, Q166-Q169].

Recommendation: Construct a deployment-specific evaluation set covering the three target document genres (subsidy notices, land-use updates, credit-scheme terms) with at least 200-500 sentences per genre, drawn from real federal/regional bureau and ACSI/ADCSI documents (with appropriate redaction).

ID	Evidence
Q161	“NLLB-MD covers the following six languages: Central Aymara (ayr_Latn), Bhojpuri (bho_Deva), Dyula (dyu_Latn), Friulian (fur_Latn), Russian (rus_Cyrl) and Wolof (wol_Latn).” (p.24)
Q166	“We translate the English side of the WMT21 English-German development set, containing a sample of newspapers from 2020 (Akhbardeh et al., 2021).” (p.25)
Q167	“We translate text extracted from a series of scripted English-language talks covering a variety of topics.” (p.25)
Q168	“We extract 3,000 utterances from the multi-session chat dataset of Xu et al. (2022), which contains on average 23 words per turn.” (p.25)
Q169	“We translated one World Health Organisation report (Donaldson and Rutter, 2017) and combined it with sentences translated from the English portion of the TAUS Corona Crisis Report.” (p.25)